

Maxwell's Relation and Thermodynamic Potential

1. Maxwell Relations

$F = F(x, y)$ and if it is perfect differential then $dF = Mdx + Ndy$

Where $M = \left(\frac{\partial F}{\partial x}\right)_y$ and $N = \left(\frac{\partial F}{\partial y}\right)_x$ then M and N will satisfy the condition

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

Maxwell relations are relationship between two derivatives of thermodynamic variables, and energy due to the equivalence of potential second derivative under a change of operation order

$\frac{d^2 F}{dx dy} = \frac{d^2 F}{dy dx}$ where F is thermodynamic potential and x and y are two of its natural

independent variables.

Maxwell relations are extremely important for two reasons.

First they show us that derivative of thermodynamic parameters are not all independent. This can serve as a consistency check in both experiments and in theoretical analysis.

Maxwell relations provide a method for expressing some derivative in other ways. This enables as to connect difficult to measure quantities to those which are readily accessible experimentally.

The measurement of entropy and chemical potential can not be directly measurable in lab but with the help of Maxwell relation there thermodynamic property can be determine theoretically.

For **Maxwell relation**.

Let us Legendre the independent variable as x , and y such that

$$U = U(x, y), S = S(x, y) \quad V = V(x, y)$$

So

$$dU = \left(\frac{\partial U}{\partial x} \right)_y dx + \left(\frac{\partial U}{\partial y} \right)_x dy$$

$$dS = \left(\frac{\partial S}{\partial x} \right)_y dx + \left(\frac{\partial S}{\partial y} \right)_x dy$$

$$dV = \left(\frac{\partial V}{\partial x} \right)_y dx + \left(\frac{\partial V}{\partial y} \right)_x dy$$

From first law of thermodynamic

$$dU = TdS - PdV$$

$$\left(\frac{\partial U}{\partial x} \right)_y = T \left(\frac{\partial S}{\partial x} \right)_y - P \left(\frac{\partial V}{\partial x} \right)_y$$

$$\left(\frac{\partial U}{\partial y} \right)_x = T \left(\frac{\partial S}{\partial y} \right)_x - P \left(\frac{\partial V}{\partial y} \right)_x$$

Hence U, V , and S are perfect differential.

Then

$$\left[\frac{\partial}{\partial y} \left(\frac{\partial U}{\partial x} \right)_y \right]_x = \left[\frac{\partial}{\partial x} \left(\frac{\partial U}{\partial y} \right)_x \right]_y$$

$$\left[\frac{\partial}{\partial y} \left(\frac{\partial V}{\partial x} \right)_y \right]_x = \left[\frac{\partial}{\partial x} \left(\frac{\partial V}{\partial y} \right)_x \right]_y$$

$$\left[\frac{\partial}{\partial y} \left(\frac{\partial S}{\partial x} \right)_y \right]_x = \left[\frac{\partial}{\partial x} \left(\frac{\partial S}{\partial y} \right)_x \right]_y$$

$$\left[\frac{\partial}{\partial y} \left(\frac{\partial U}{\partial x} \right)_y \right]_x = \left(\frac{\partial T}{\partial y} \right)_x \left(\frac{\partial S}{\partial x} \right)_y + T \left[\frac{\partial}{\partial y} \left(\frac{\partial S}{\partial x} \right)_y \right]_x - \left(\frac{\partial P}{\partial y} \right)_x \left(\frac{\partial V}{\partial x} \right)_y - P \left[\frac{\partial}{\partial y} \left(\frac{\partial V}{\partial x} \right)_y \right]_x \quad \text{---(1)}$$

Similarly

$$\left[\frac{\partial}{\partial x} \left(\frac{\partial U}{\partial y} \right)_x \right]_y = \left(\frac{\partial T}{\partial x} \right)_y \left(\frac{\partial S}{\partial y} \right)_x + T \left[\frac{\partial}{\partial x} \left(\frac{\partial S}{\partial y} \right)_x \right]_y - \left(\frac{\partial P}{\partial x} \right)_y \left(\frac{\partial V}{\partial y} \right)_x - P \left[\frac{\partial}{\partial x} \left(\frac{\partial V}{\partial y} \right)_x \right]_y \quad \text{---(2)}$$

Equating equation (1) and (2)

$$\left(\frac{\partial P}{\partial x} \right)_x \left(\frac{\partial V}{\partial y} \right)_x - \left(\frac{\partial P}{\partial y} \right)_x \left(\frac{\partial V}{\partial x} \right)_y = \left(\frac{\partial T}{\partial x} \right)_y \left(\frac{\partial S}{\partial y} \right)_x - \left(\frac{\partial T}{\partial y} \right)_x \left(\frac{\partial S}{\partial x} \right)_y \quad \text{(A)}$$

Maxwell first relation:- put $x = T, y = V$

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

Maxwell Second Relation:- put $x = T, y = P$

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$$

Maxwell Third Relation:- put $x = S, y = V$

$$\left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

Maxwell Fourth Relation:- put $x = S, y = P$

$$\left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_P$$

Thermodynamic potential is a scalar function used to represent the thermodynamic state of system. The concept of thermodynamic potentials was introduced by Pierre Duhem in 1886.

One main thermodynamic potential that has a physical interpretation is the internal energy. It is energy of configuration of a given system of conservative forces. Expression for all other thermodynamic energy potentials are derivable via Legendre transformation.